

Solving PDEs on a Quantum Computer

HHL, the variational linear solver, and ColibriTD's H-DES

Lecture 9

October 3, 2026

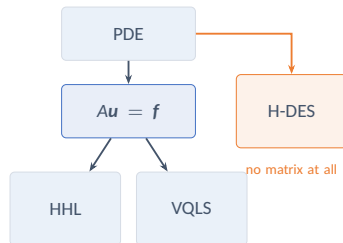
Where we are

L8 left us with

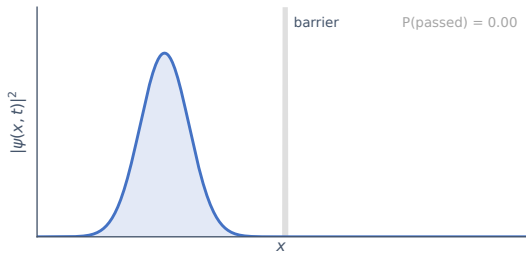
- A PDE, discretized, is a sparse linear system $A\mathbf{u} = \mathbf{f}$
- Our rod: $-u'' = 1$, $N = 4$, answer (0.08, 0.12, 0.12, 0.08)
- N numbers fit in $\log_2 N$ qubits as amplitudes

Today: three quantum routes

- 1 **HHL** — invert A coherently (fault-tolerant era)
- 2 **VQLS** — search for $|u\rangle$ variationally (NISQ)
- 3 **H-DES** — skip A entirely; learn the function $u(x)$ (NISQ)



A quantum computer already solves one PDE



a wave packet meets a barrier: part reflects, part tunnels

Schrödinger's equation is a *linear* PDE:

$$i\hbar \psi_t = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi$$

- Solution: $|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle$, a *unitary*
- Applying unitaries is what a quantum computer does natively

Everything else needs a trick

- Poisson, heat: not unitary
- Burgers, Navier–Stokes: nonlinear

The quantum linear systems problem

QLSP

Given an $N \times N$ Hermitian matrix A (s -sparse, condition number κ) and a circuit that prepares $|b\rangle$,
output a state $|x\rangle$ within distance ϵ of $\frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}$.

Our rod as a QLSP ($N = 4, n = 2$)

$$|b\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$|x\rangle = 0.39 |00\rangle + 0.59 |01\rangle \\ + 0.59 |10\rangle + 0.39 |11\rangle$$

Read the fine print

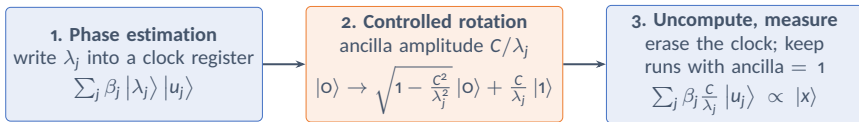
- Input: $|b\rangle$ must be cheap to prepare
- Output: a *state* — the scale $\|x\|$ is lost, and reading all N amplitudes costs $\mathcal{O}(N)$ shots
- Useful outputs: $\langle x|M|x\rangle$, e.g. the average temperature

The HHL idea: divide by eigenvalues

Expand $|b\rangle$ in the eigenvectors of A ($A|u_j\rangle = \lambda_j|u_j\rangle$):

$$|b\rangle = \sum_j \beta_j |u_j\rangle \quad \Rightarrow \quad |x\rangle \propto A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

So we need a circuit that multiplies each eigen-component by $1/\lambda_j$. Three steps:



Phase estimation needs $U = e^{iAt}$ and its powers, each *controlled* by a clock qubit. $C \leq \lambda_{\min}$ keeps the rotation valid.

—
A. W. Harrow, A. Hassidim, S. Lloyd (2009). "Quantum algorithm for linear systems of equations." *Phys. Rev. Lett.* 103, 150502.
arXiv:0811.3171

Worked HHL: a two-point rod with a point load

Poisson with $N = 2$ (multiply by h^2) and a unit load at the first node:

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad |b\rangle = |0\rangle, \quad \text{classical answer } \mathbf{x} = A^{-1}\mathbf{b} = \left(\frac{2}{3}, \frac{1}{3}\right).$$

Eigen-decomposition

- $\lambda = 1$: $|+\rangle$ $\lambda = 3$: $|-\rangle$
- $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$
- Target: $|x\rangle \propto \frac{1}{4}|+\rangle + \frac{1}{3}|-\rangle \propto 2|0\rangle + |1\rangle$

Choosing t so the clock is exact

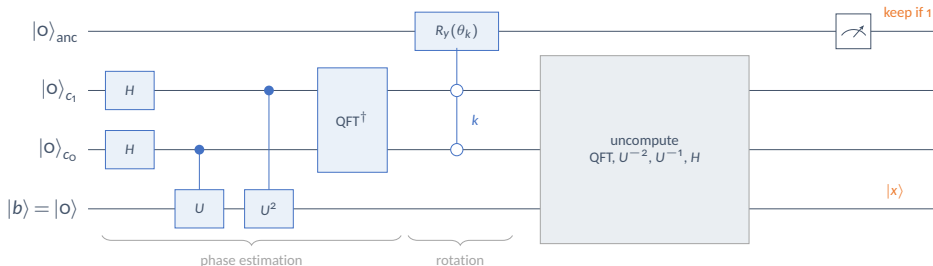
- Two clock qubits; pick $t = \pi/2$:

$$e^{i\lambda t} = e^{2\pi i \lambda/4}$$

- $\lambda = 1 \rightarrow |01\rangle, \lambda = 3 \rightarrow |11\rangle$
- And $U = e^{iA\pi/2} = iX, U^2 = -\mathbb{I}$

Takeaway: $U = iX$ means controlled- U is just an S gate plus a CNOT, and controlled- U^2 is a single Z . The whole circuit is a handful of gates.

The HHL circuit for this example



Gate by gate

- controlled- $U = S$ on c_0 , then CNOT $c_0 \rightarrow s$
- controlled- $U^2 = Z$ on c_1

Rotation angles ($C = 1$, $\theta_k = 2 \arcsin(C/k)$)

- clock $k = 1$ ($|01\rangle$): $\theta = \pi$
- clock $k = 3$ ($|11\rangle$): $\theta = 2 \arcsin \frac{1}{3} = 0.680$

Tracing the state

Order: ancilla, clock, system. Statevector-simulated; every line below is exact.

After	State
phase estimation	$\frac{1}{\sqrt{2}} (0\rangle 01\rangle +\rangle + 0\rangle 11\rangle -\rangle)$ eigenvalue now readable in the clock
rotation	$\frac{1}{\sqrt{2}} (1\rangle 01\rangle +\rangle + (\sqrt{\frac{8}{9}} 0\rangle + \frac{1}{3} 1\rangle) 11\rangle -\rangle)$
uncompute	$ 1\rangle 00\rangle (\frac{2}{3} 0\rangle + \frac{1}{3} 1\rangle) + 0\rangle 00\rangle (\dots)$

Measure the ancilla, get 1

System: $\frac{1}{\sqrt{5}} (2|0\rangle + |1\rangle) = (0.894, 0.447)$

Classical: $(\frac{2}{3}, \frac{1}{3}) / \|\cdot\| = (0.894, 0.447) \checkmark$

- Success probability $\frac{4}{9} + \frac{1}{9} = \frac{5}{9}$
- The amplitudes $(\frac{2}{3}, \frac{1}{3})$ even carry the true scale because $C = 1$
- Failure $\Rightarrow$ rerun; amplitude amplification cuts the repeats

What HHL costs — and the fine print

Runtime (sparse A , precision ϵ)

HHL 2009 $\mathcal{O}(\log N \cdot s^2 \kappa^2 / \epsilon)$

CKS 2017 $\mathcal{O}(s \kappa \cdot \text{polylog}(s \kappa / \epsilon))$

Costa et al. 2022 $\mathcal{O}(\kappa \log(1/\epsilon))$ queries — optimal

Exponentially better than $\mathcal{O}(N)$ in the size — if κ stays small.

Why not on today's hardware

- Phase estimation: long, coherent, controlled e^{iAt}
- Needs fault tolerance (L12–L16)

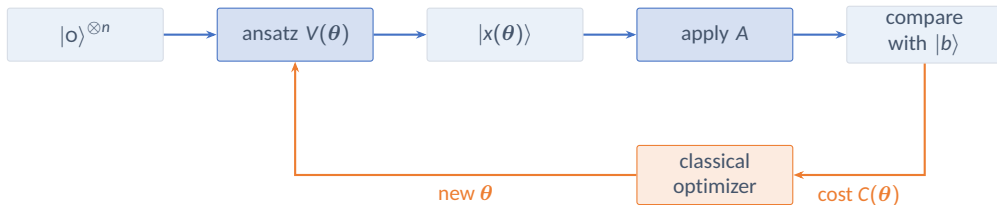
The fine print for PDEs

- **Conditioning:** Poisson has $\kappa \sim N^{2/d}$. In 1D, $\kappa \sim N^2$ wipes out the gain
- **Input:** loading a general f into $|b\rangle$ can cost $\mathcal{O}(N)$
- **Output:** a state, not the field
- **Dimension:** for finite elements in fixed low d , the speedup is at most polynomial; the promise lives in high d

Childs, Kothari, Somma, *SIAM J. Comput.* 46, 1920 (2017) · Costa et al., *PRX Quantum* 3, 040303 (2022) · Aaronson, "Read the fine print," *Nature Physics* 11, 291 (2015) · Montanaro & Pallister, *Phys. Rev. A* 93, 032324 (2016)

The VQLS idea

Do not invert A . Search for the state that A maps onto $|b\rangle$ — the VQE loop of L6 with a new cost.



- Circuit depth is set by the ansatz, which we choose — not by κ or $1/\epsilon$
- The price: no guarantee. Trainability, barren plateaus and shot noise from L5 all apply

C. Bravo-Prieto, R. LaRose, M. Cerezo, Y. Subasi, L. Cincio, P. J. Coles. “Variational Quantum Linear Solver.” arXiv:1909.05820 (2019); *Quantum* 7, 1188 (2023).

The cost function: global and local

Let $|\Psi(\theta)\rangle = A V(\theta) |o\rangle$ and $|b\rangle = U_b |o\rangle$. We want $|\Psi\rangle \parallel |b\rangle$.

Global cost

$$C_G(\theta) = 1 - \frac{|\langle b | \Psi \rangle|^2}{\langle \Psi | \Psi \rangle}$$

- $C_G = 0 \iff A|x\rangle \propto |b\rangle$
- The denominator removes the scale — $|x\rangle$ is normalized anyway
- **Barren plateau**: gradients vanish exponentially in n

Local cost

$$C_L(\theta) = 1 - \frac{1}{n} \sum_{j=1}^n \frac{\langle \Psi | U_b (|o\rangle\langle o|_j \otimes \mathbb{I}) U_b^\dagger | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

- One-qubit projectors: checks each qubit separately
- Same zero set, and $C_L \leq C_G \leq n C_L$
- Trainable for shallow ansätze

Optimize C_L ; report C_G (or the fidelity).

Step 1: write A as a sum of unitaries

Hardware applies unitaries; A is not one. Write $A = \sum_{\ell} c_{\ell} A_{\ell}$ with each A_{ℓ} unitary.

Our rod matrix, two qubits (index $|q_1 q_0\rangle$)

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} = 2\mathbb{I}\mathbb{I} - \mathbb{I}X - \frac{1}{2}XX - \frac{1}{2}YY$$

- $\mathbb{I}X$ flips q_0 : links $00 \leftrightarrow 01$ and $10 \leftrightarrow 11$
- $\frac{1}{2}(XX + YY)$ swaps $01 \leftrightarrow 10$; the $00 \leftrightarrow 11$ terms cancel

$L = 4$ Pauli terms

Each Pauli string is unitary, so this is a valid decomposition — checked exactly.

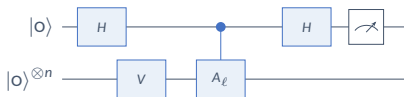
Why L matters

- Each cost evaluation needs $\mathcal{O}(L^2)$ overlaps
- A blind Pauli expansion can need up to 4^n terms
- Exploit structure: a banded A stays $\mathcal{O}(1)$ shift operators

Step 2: measure the cost with Hadamard tests

Expand C_G : every term is an overlap built from V , U_b and the A_ℓ :

$$\langle \Psi | \Psi \rangle = \sum_{\ell, \ell'} c_\ell c_{\ell'}^* \langle o | V^\dagger A_{\ell'}^\dagger A_\ell V | o \rangle, \quad \langle b | \Psi \rangle = \sum_{\ell} c_\ell \langle o | U_b^\dagger A_\ell V | o \rangle.$$



$\langle Z_{\text{anc}} \rangle = \text{Re} \langle o | V^\dagger A_\ell V | o \rangle$; add $S^\dagger$ for the imaginary part

Counting for our rod ($L = 4$)

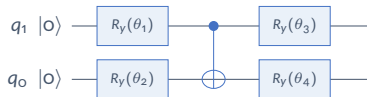
- $\langle b | \Psi \rangle$: 4 overlaps
- $\langle \Psi | \Psi \rangle$: $\binom{4}{2} = 6$ cross terms (diagonal ones equal 1)
- Each estimated to ϵ with $\mathcal{O}(1/\epsilon^2)$ shots — *per iteration*

Takeaway: Shots, not qubits, are what make VQLS expensive: $\mathcal{O}(L^2/\epsilon^2)$ per cost evaluation, times hundreds of optimizer iterations.

Worked VQLS: the heated rod on two qubits

Same system as L8 ($N = 4$). Uniform heating means $\mathbf{f} \propto (1, 1, 1, 1)$, so $|b\rangle = |++\rangle = (H \otimes H) |00\rangle$.

Ansatz — 4 parameters, real amplitudes



Target

$$|x\rangle = (0.392, 0.588, 0.588, 0.392)$$

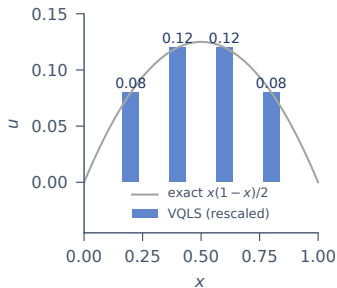
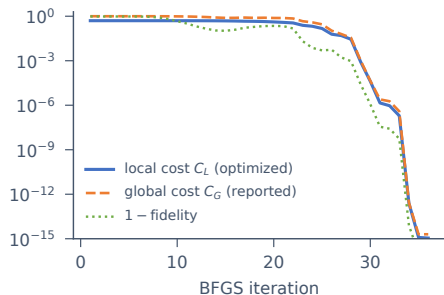
Note $|\langle b|x\rangle|^2 = 0.96$: a good guess is not yet a solution.

Recipe

- 1 Random start θ_0
- 2 Evaluate $C_L(\theta)$ (Hadamard tests)
- 3 BFGS updates θ
- 4 Stop when $C_L < 10^{-10}$
- 5 Recover the scale with one more overlap:

$$\mathbf{x} = \frac{\|\mathbf{f}\|}{\langle b|A|x\rangle} |x\rangle$$

Worked VQLS: result



- 36 BFGS iterations (exact statevector); fidelity $1 - 10^{-15}$; rescaled answer (0.08, 0.12, 0.12, 0.08) — the Thomas result from L8
- Note the first ~ 15 iterations: C_L sits near 0.5 — a small plateau even at $n = 2$
- $C_L \leq C_G$ throughout, as the bound promises

The catch: with real shots each point on this curve costs thousands of circuit runs, and the cost cannot be resolved below the shot-noise floor $\sim L/\sqrt{\text{shots}}$.

HHL against VQLS

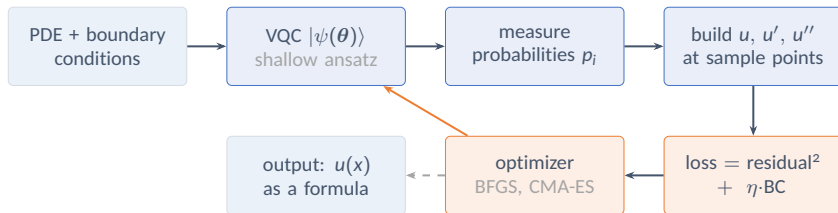
	HHL	VQLS
Era	fault-tolerant	NISQ
Depth	grows with κ and $1/\epsilon$	fixed by the ansatz
Guarantee	provable runtime	none — heuristic
Building blocks	phase estimation, controlled rotations	ansatz, Hadamard tests, optimizer
Fails when	hardware is not fault-tolerant	ansatz misses $ x\rangle$, barren plateaus, shot noise
Output	$ x\rangle$, a quantum state	$ x\rangle$, a quantum state
Our examples	$N = 2$, exact, success 5/9	$N = 4$, fidelity $1 - 10^{-15}$

- Both inherit the same pipeline: *grid* $\rightarrow$ *matrix* $\rightarrow$ *state*
- Both return amplitudes on a grid — to see the field you must sample it $\mathcal{O}(N)$ times
- Both need a **linear** PDE. A nonlinear one means many solves (Newton) or a linearization

Takeaway: Can we avoid the matrix, the grid readout, *and* the linearity requirement? That is the design goal of H-DES.

H-DES: learn the function, not the vector

Instead of N grid values, represent $u(x)$ by a few **Chebyshev polynomials** $T_k(x)$ — a spectral method (L8) — and let a quantum circuit supply the coefficients.



- No discretization matrix, no $|b\rangle$ to load, no grid to read out
- The PDE enters only in the *classical* loss — so it may be **nonlinear**

H. Jaffali, J. Bastos de Araujo, N. Milazzo, M. Reina, H. de Boutray, K. Baumann, F. Holweck, Y. Mohdeb, R. Katz (ColibriTD). "H-DES: a Quantum-Classical Hybrid Differential Equation Solver." arXiv:2410.01130 (v1 2024, v2 2025).

The encoding: probabilities become coefficients

n qubits = one **sign qubit** + $(n - 1)$ **index qubits** $\Rightarrow 2^{n-1}$ Chebyshev polynomials.

Basis-state encoding

$$u(x) = \lambda \sum_{i=0}^{2^{n-1}-1} (p_i - p_{i+2^{n-1}}) T_i(x), \quad p_i = |\langle i | \psi(\theta) \rangle|^2, \quad \lambda \in \mathbb{R} \text{ trained}$$

As an observable

$$u(x) = \lambda \langle \psi | O(x) | \psi \rangle$$

$$O(x) = Z \otimes \sum_i T_i(x) |i\rangle \langle i|$$

- Z on the sign qubit makes negative coefficients possible
- $O(x)$ is *diagonal*: one set of computational-basis shots gives u at every x

Derivatives are exact

$$u^{(q)}(x) = \lambda \sum_i (p_i - p_{i+2^{n-1}}) T_i^{(q)}(x)$$

- Same p_i , just differentiate the known polynomials — no finite differences, no extra circuits
- One more qubit doubles the basis

The loss and the loop

Pick sample (collocation) points x_s . For a PDE $\mathcal{R}[u] = 0$ with boundary data $u(x_B) = g_B$:

$$\mathcal{L}(\theta, \lambda) = \underbrace{\frac{1}{n_s} \sum_s \mathcal{R}[u](x_s)^2}_{\text{does } u \text{ satisfy the PDE?}} + \eta \underbrace{\frac{1}{n_B} \sum_B (u(x_B) - g_B)^2}_{\text{does it meet the boundary?}}$$

Design choices in the paper

- Hardware-efficient ansatz: R_y layers + linear CNOTs, depth $d = 2-5$
- R_y only: probabilities need no complex amplitudes
- BFGS on emulators; SLSQP or CMA-ES on hardware
- Systems of equations: one circuit per unknown function

Two refinements

- *Floating boundary conditions*: shift u by a polynomial so the BCs hold exactly, and drop the penalty
- *Pauli-monomial encoding*:
 $O(x) = \sum_i o_i T_i(x) P_i$ with k -local Paulis — fewer measurements, milder barren plateaus

Worked H-DES: the same heated rod

Solve $-u'' = 1$, $u(0) = u(1) = 0$ with $n = 3$ qubits $\Rightarrow T_0, \dots, T_3$ on $s = 2x - 1$.

What the answer must look like

$$u = \frac{1}{2}x(1-x) = \frac{1}{16}(T_0 - T_2)$$

so the circuit must learn

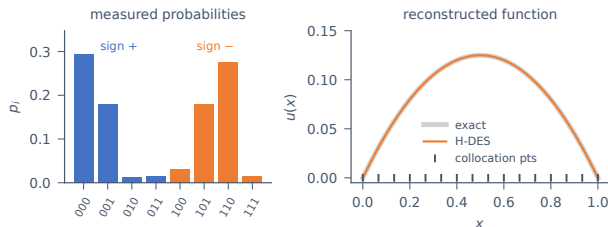
$$\lambda(p_0 - p_4) = \frac{1}{16}, \quad \lambda(p_2 - p_6) = -\frac{1}{16}$$

and zero for T_1, T_3 .

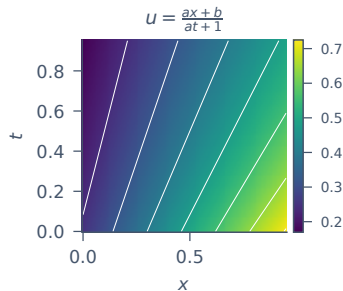
Set-up

- depth 2: 9 angles + $\lambda = 10$ parameters
 - 16 sample points, $\eta = 10$, BFGS
 - $u'' = 4\lambda \sum(\dots) T_i''(s)$ (chain rule)
- Loss 3×10^{-15} ; learned coefficients (0.0625, 0, -0.0625, 0); max error 2×10^{-8}
 - Reading the p_i with finite shots: max error 8.6×10^{-3} (10^3), 2.9×10^{-3} (10^4), 8.7×10^{-4} (10^5)

Our re-implementation of the published method in a statevector emulator; the map $x \mapsto 2x - 1$ is our choice.



H-DES on real hardware: a nonlinear PDE



the exact target, $a = 0.5$, $b = 0.25$

Inviscid Burgers — the nonlinear equation of L8:

$$u_t + u u_x = 0, \quad u(x, 0) = ax + b, \quad u = \frac{ax + b}{at + 1}$$

- Two variables: qubits split between x and t ; tensor-product Chebyshev basis
- IBM Heron r2 processors (`ibm_fez`, `ibm_marrakesh`), 156 qubits
- *Circuit stacking*: identical copies on disjoint qubits (a 40-qubit stacked circuit) to cut shot cost
- CMA-ES with a staged shot budget, $500 \rightarrow 10,000$ shots
- Best loss 1.8×10^{-3} ; no explicit error mitigation

Also solved: a nonlinear 1D hypoelastic tensile test (15 + 4 qubits).
The method ships as IBM's *QUICK-PDE* Qiskit Function.

K. Baumann, Y. Mohdeb, R. Randrianarisoa, R. Katz, A. Boyle, F. Holweck (2026). "Solving nonlinear differential equations on noisy 156-qubit quantum computers." arXiv:2601.04439 · IBM Quantum docs, "QUICK-PDE: A Qiskit Function by ColibriTD."

H-DES, assessed honestly

Strengths

- Few qubits: the basis doubles per qubit
- Exact derivatives from the same measurements
- Nonlinear PDEs at no extra quantum cost — the nonlinearity lives in the classical loss
- Shallow circuits; already run on 156-qubit hardware
- Output is a formula: evaluate u anywhere

Open questions

- Basis-state readout estimates 2^n probabilities: shots grow exponentially for large n
- Barren plateaus for global diagonal observables
- No proven advantage; demonstrated solutions are smooth, low-degree functions that classical spectral methods solve in milliseconds
- Heuristic optimizer, random initialization

Takeaway: H-DES is best read as a *quantum-parameterized spectral method*: a compact, trainable model of $u(x)$ — not a quantum speedup for linear algebra.

Three quantum solvers, one rod

	HHL	VQLS	H-DES
Works on	$Au = f$	$Au = f$	the PDE itself
Quantum state holds	grid values (amplitudes)	grid values (amplitudes)	spectral coefficients (probabilities)
Qubits	$\log_2 N + \text{clock} + 1$	$\log_2 N$	$1 + \log_2(\#\text{basis})$
Nonlinear PDEs	via linearization	via linearization	directly
Guarantee	provable	heuristic	heuristic
Hardware era	fault-tolerant	NISQ	NISQ (shown on 156 qubits)
Our example	$N = 2$, exact	$N = 4$, 2 qubits, exact	3 qubits, error 2×10^{-8}

How to read a paper in this area: check whether state preparation, readout and shot cost are counted. Then check the classical baseline: for Poisson it is linear-time multigrid.

Programming Assignment 2

Task

Solve the 1D Poisson equation with a source of your choice on $n = 3$ and $n = 4$ qubits with VQLS, and characterize the solver.

Deliverables

- 1 Working notebook: problem setup, decomposition of A , ansatz, local cost, optimizer.
- 2 Convergence comparison across at least two optimizers.
- 3 Shot-noise study at three shot budgets, with the noise floor identified.
- 4 Gradient-variance study: global vs local cost, at two depths.
- 5 A two-page report stating what you measured and what you conclude — including a clear statement of what your results do *not* show.

Grading — correctness 40%, experimental design 25%, analysis and honesty 25%, presentation 10%.

Summary

- The QLSP: from A and $|b\rangle$, produce $|x\rangle \propto A^{-1} |b\rangle$ — a state, not a vector
- **HHL** divides by eigenvalues via phase estimation; worked 2×2 Poisson:
 $|x\rangle = (2|0\rangle + |1\rangle)/\sqrt{5}$ with probability $5/9$
- **VQLS** searches with a shallow ansatz and the local cost; our rod in 2 qubits,
 $A = 2\mathbb{I}\mathbb{I} - \mathbb{I}X - \frac{1}{2}XX - \frac{1}{2}YY$
- **H-DES** encodes $u(x)$ in Chebyshev coefficients read from probabilities; exact derivatives; nonlinear PDEs handled in the classical loss
- None has a proven end-to-end advantage for a practical PDE yet

Next — L10: quantum information theory: density matrices, partial trace, fidelity.

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Variational and nonlinear differential-equation solvers

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